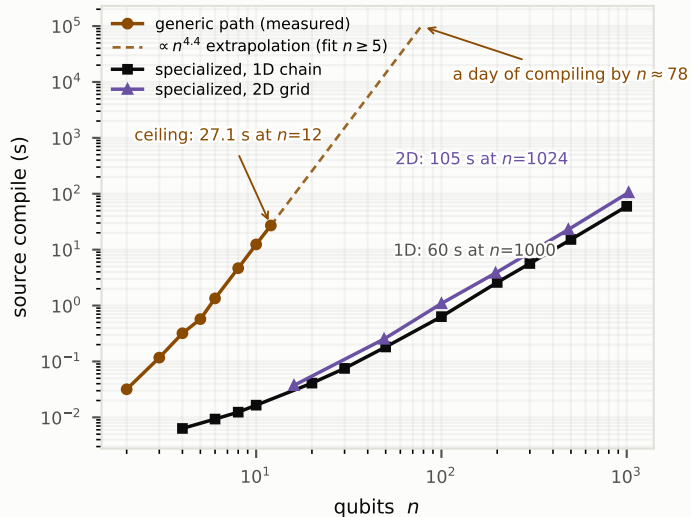
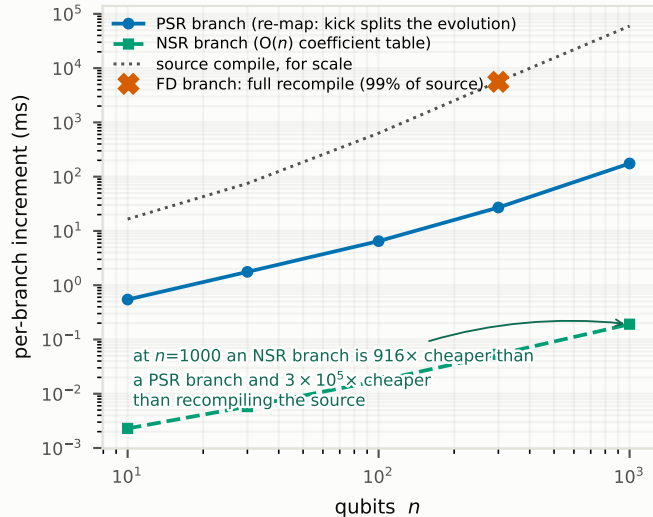


(a) compiling the source program



(b) the price of differentiating it



wall clock on macOS-26.5-arm64-arm-64bit; schedule ops + pulse ledger only (no pulse-shape synthesis). 2D grid: compiled NN model exact, diagonal $J/8$ tail ($\sim 14\%$ L1) declared-dropped.